WORKED SOLUTION

STEP III 2012, Question 2

Pure • Infinite Products and Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q2

The key idea

The product in (i) collapses one bracket at a time through the difference of two squares, so it telescopes to something very simple. Once $\frac{1}{1-x}$ is written as an infinite product, taking logarithms turns the product into a sum, and differentiating that sum term by term gives the series of fractions. Part (ii) is the same trick in disguise: $1+x+x^2$ is a quotient of two expressions of the form $1-t$, so the result of (i) applies to each of them.

Solution

Throughout, $|x|<1$ and, as the question allows, we don't worry about convergence.

Part (i)

Combine the brackets from the left, using $(1-a)(1+a)=1-a^2$ each time:

$$\begin{aligned}(1-x)(1+x)&=1-x^2,\\(1-x^2)(1+x^2)&=1-x^4,\\(1-x^4)(1+x^4)&=1-x^8,\end{aligned}$$

and so on. Each step doubles the power, and the last bracket $1+x^{2^n}$ turns $1-x^{2^n}$ into $1-x^{2^{n+1}}$. So

$$(1-x)(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n})=1-x^{2^{n+1}}.$$

Since $x\neq 1$ we can divide by $1-x$:

$$(1+x)(1+x^2)\cdots(1+x^{2^n})=\frac{1}{1-x}-\frac{x^{2^{n+1}}}{1-x},$$

which rearranges to the required expression for $\frac{1}{1-x}$.

Now let $n \to \infty$. Because $|x| < 1$, $x^{2^{n+1}} \to 0$, so the remainder term vanishes and

$$\frac{1}{1-x} = \prod_{r=0}^{\infty}(1 + x^{2^r}).$$

Every factor is positive (as $x^{2^r} > -1$), and so is $1 - x$, so we may take logarithms. The log of a product is the sum of the logs:

$$-\ln(1-x) = \sum_{r=0}^{\infty} \ln(1 + x^{2^r}),$$

which is the required result after multiplying by -1.

Ask yourself How do we get from a sum of logarithms to a sum of fractions?

Differentiate both sides with respect to x. The derivative of $\ln(1 + x^{2^r})$ is, by the chain rule,

$$\frac{2^r x^{2^r - 1}}{1 + x^{2^r}}.$$

On the left, $\frac{\mathrm{d}}{\mathrm{d}x}(-\ln(1-x)) = \frac{1}{1-x}$. Differentiating term by term (the convergence issue we are told to ignore) gives

$$\frac{1}{1-x} = \sum_{r=0}^{\infty} \frac{2^r x^{2^r-1}}{1 + x^{2^r}} = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \cdots,$$

where the terms shown are $r = 0, 1, 2$. This is the required series.

Part (ii)

The left-hand side is the derivative of $\ln(1 + x + x^2)$, since $\frac{\mathrm{d}}{\mathrm{d}x}(1 + x + x^2) = 1 + 2x$. So we look for a log-sum for $1 + x + x^2$, and the factorisation $1 - x^3 = (1-x)(1 + x + x^2)$ gives one:

$$\ln(1 + x + x^2) = \ln(1 - x^3) - \ln(1 - x).$$

Since $|x^3| < 1$ too, the log result of (i) holds with x replaced by x^3. Writing $y_r = x^{2^r}$, so that $(x^3)^{2^r} = y_r^3$,

$$\ln(1 + x + x^2) = -\sum_{r=0}^{\infty} \ln(1 + y_r^3) + \sum_{r=0}^{\infty} \ln(1 + y_r) = -\sum_{r=0}^{\infty} \ln \frac{1 + y_r^3}{1 + y_r}.$$

The sum of cubes factorises as $1 + y^3 = (1+y)(1 - y + y^2)$, so each fraction is $1 - y_r + y_r^2$ and

$$\ln(1 + x + x^2) = -\sum_{r=0}^{\infty} \ln(1 - x^{2^r} + x^{2^{r+1}}).$$

(Every $1 - y + y^2 = (y - \frac{1}{2})^2 + \frac{3}{4}$ is positive, so these logarithms make sense.)

Ask yourself Where does the minus sign go when we differentiate?

The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(1-x^{2^r}+x^{2^{r+1}}) = \frac{-2^r x^{2^r-1}+2^{r+1}x^{2^{r+1}-1}}{1-x^{2^r}+x^{2^{r+1}}} = -\frac{2^r x^{2^r-1}(1-2x^{2^r})}{1-x^{2^r}+x^{2^{r+1}}},$$

and this minus sign cancels the one in front of the sum. Differentiating both sides term by term,

$$\frac{1+2x}{1+x+x^2} = \sum_{r=0}^{\infty}\frac{2^r x^{2^r-1}(1-2x^{2^r})}{1-x^{2^r}+x^{2^{r+1}}}.$$

The first three terms are

$$\begin{aligned} r=0: &\quad \frac{1-2x}{1-x+x^2},\\ r=1: &\quad \frac{2x(1-2x^2)}{1-x^2+x^4} = \frac{2x-4x^3}{1-x^2+x^4},\\ r=2: &\quad \frac{4x^3(1-2x^4)}{1-x^4+x^8} = \frac{4x^3-8x^7}{1-x^4+x^8}, \end{aligned}$$

which is exactly the series we were asked to establish.

Another route. The identity $(1+y+y^2)(1-y+y^2) = 1+y^2+y^4$ gives a telescoping product for $1+x+x^2$ directly, just like the one in (i): $1+x+x^2 = (1+x^{2^{n+1}}+x^{2^{n+2}})/\prod_{r=0}^{n}(1-x^{2^r}+x^{2^{r+1}})$, and the numerator tends to 1.

Checks

- Put $x=0$: both sides of each series equal 1.
- At $x=\frac{1}{2}$ the left-hand side of (ii) is $\frac{2}{7/4}=\frac{8}{7}\approx 1.142857$. The first four terms of the right-hand side sum to 1.142369, and the error shrinks very fast because the powers x^{2^r} do.
- We checked the telescoped product exactly with fractions, and both series and the log identities numerically at 1000 random x in $(-0.95, 0.95)$, using 30 terms.

Common slips

- Losing a factor in the chain rule: the derivative of x^{2^r} is $2^r x^{2^r-1}$, not $2^r x^{2^r}$.
- Getting the overall sign wrong in (ii): the derivative of $1-y+y^2$ is $-1+2y$, and it's this minus that cancels the minus in front of the sum.

Beyond STEP

The product $\prod(1 + x^{2^r})$ says that every positive integer can be written in exactly one way as a sum of distinct powers of 2, which is binary notation expressed as a generating function. "Ignoring convergence" hides real questions: when may an infinite product be logged, and when may a series be differentiated term by term? A first analysis course answers these with uniform convergence, and on $|x| \leq c < 1$ everything here is uniformly convergent. See Abbott, *Understanding Analysis*, for a clear treatment.

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